Barriers to AI tool adoption

"What barriers prevent non-technical users and small business owners from discovering, selecting, and effectively using appropriate AI tools?" *Generated by LeapSpace*

Barriers to Discovering, Selecting, and Using AI Tools at a Glance

Key Findings Table

Barrier stage	Main barriers	Most affected groups/settings	Evidence
Discovery	Crowded markets, weak use-case clarity, lack of direct comparison signals, and overreliance on peers or vendors because users cannot evaluate tools technically	Non-technical users, SME owners, first-time AI adopters	[1] [2] [3] [4] [5]
Selection	Trust, transparency, fairness, and black-box concerns; difficulty judging accuracy, reliability, and privacy risk; unclear total cost and workflow fit	Users making consequential business decisions; firms lacking internal expertise	[6] [7] [8] [9] [10] [11] [12] [13] [14] [15]
Implementation	Poor data readiness, fragmented data, legacy systems, weak interoperability, and lack of technical staff	SMEs and micro/small firms	[16] [17] [18] [19] [20] [21] [22] [23]
Effective use	Steep learning curves, unclear interfaces, weak onboarding, insufficient training, and limited time to learn	Non-technical end users in resource-constrained environments	[24] [25] [26] [27] [28] [29] [30]
Sustained use	Workflow mismatch, productivity loss during integration, hidden personnel costs, and privacy/compliance uncertainty	Small firms embedding AI into daily operations	[14] [31] [32] [33] [34] [35] [36] [37] [38]

Barrier stage	Main barriers	Most affected groups/settings	Evidence
Support ecosystem	Success depends heavily on vendor training, support, and low-code/no-code interfaces, but these supports do not remove trust or workflow-fit issues	Non-expert users and SMEs without in-house AI capability	[29] [30] [39] [40] [41] [42]

Direct Answer

The main barriers form a sequence.

1. **Before adoption**, non-technical users and small business owners struggle to **discover** suitable AI tools because the market is crowded, use cases are unclear, and the evidence base says little about how they actually search and shortlist tools in real marketplaces; in practice they often fall back on peer advice, vendor messaging, and social influence rather than systematic evaluation [1] [2] [4] [5] [43] [44] [45] [46].

2. **At selection**, they struggle to **judge** whether a tool is trustworthy, accurate, fair, private, and worth the effort because AI often appears opaque; users weigh transparency, legitimacy, explainability, and reliability, but usually lack strong comparison signals or the expertise to interpret technical claims [6] [7] [8] [9] [10] [11] [12] [13] [47] [48].

3. **After choosing a tool**, the biggest barriers are usually not just subscription price but **capability and readiness gaps**: poor data quality, fragmented systems, lack of technical staff, steep learning curves, weak onboarding, integration burden, and unresolved privacy/compliance concerns [16] [17] [18] [19] [20] [22] [24] [34] [35] [37] [49] [50].

Taken together, the practical bottleneck is often less “which AI tool is best” and more “which tool is simple enough, trustworthy enough, and well-supported enough for this specific workflow” [12] [25] [30].
Confidence level: High - this conclusion is directly supported by converging evidence across the meta-synthesis and multiple topic-specific findings [1] [5] [16] [22] [24] [49].

Study Scope

- **Timeframe covered:** mainly 2018-2025, with emphasis on 2019-2025 studies in AI adoption, usability, and decision-making [1] [10] [15] [48].
- **Disciplines/settings:** SME adoption, information systems, human-computer interaction, decision support, education, healthcare, public services, and consumer AI use [29] [47] [49] [51] [52].

- **Methods represented:** surveys, reviews, qualitative interviews, mixed methods, experiments, and sector-specific implementation studies; direct behavioral studies of AI marketplace search are notably scarce [1] [11] [43] [47] [53].
  Confidence level: High - the methodological pattern is explicitly stated in the synthesis [1] [10] [54].

Assumptions & Limitations

- The strongest evidence concerns **adoption barriers and post-adoption use**, not actual marketplace discovery behavior [1] [2] [43] [44].
- No retrieved study directly compares **small business owners versus technical buyers** using the same design for AI tool selection criteria [10] [11] [55].
- Some conclusions are cross-context syntheses from healthcare, education, consumer, and organizational studies rather than head-to-head SME marketplace studies [29] [31] [41].
  Confidence level: High - these are explicit evidence gaps in the provided synthesis [1] [55].

Suggested Further Research

- Behavioral studies of how non-technical users and SMEs search, shortlist, and compare AI tools in live marketplaces [1] [44].
- Experiments on whether reviews, demos, certifications, pricing transparency, and explainability labels improve non-expert tool choice [10] [11].
- Direct comparisons of selection criteria across SME owners, managers, and technical buyers [55].
- Longitudinal work separating pre-adoption discovery barriers from implementation and sustained-use barriers [2] [34].
- Better cost evidence on hidden personnel, workflow redesign, and training burdens in small firms [14] [56].

Full Report

Foundational Context

The adoption problem here is best understood as a **three-stage pathway**: **discovery**, **selection**, and **effective use**. Each stage creates different barriers, and the literature is stronger on later stages than on the first [1] [2] [43] [44].

- **Discovery** means identifying relevant use cases and narrowing the tool set to plausible options. The evidence base is thin here, especially for real-world AI marketplaces [1] [53] [57].
- **Selection** means deciding whether a tool is trustworthy, useful, fair, easy enough to use, and appropriate for business constraints such as privacy or compliance [10] [11] [12].
- **Effective use** means getting value after adoption through onboarding, configuration, workflow integration, staff support, and ongoing governance [24] [26] [34].

A second foundational distinction is between **product-fit barriers** and **organizational-capacity barriers**. Product-fit barriers include poor interoperability, confusing interfaces, unreliable outputs, and workflow mismatch. Organizational-capacity barriers include weak data readiness, skills shortages, limited time, lack of support, and insufficient governance [17] [22] [23] [58]. The evidence suggests these are distinct but reinforcing: even a good product fails in a weak organizational environment, while a capable firm still struggles with badly fitting tools [22] [25] [31].
Confidence level: High - this framework is explicitly supported by the meta-analysis and implementation studies [17] [22].

Introduction: Why AI Tool Adoption Is Hard for Non-Experts

Across SMEs and non-technical users, barriers are not limited to software affordability. Studies repeatedly identify resource constraints, weak IT and data infrastructure, skill shortages, employee resistance, and broader institutional constraints as core inhibitors of AI adoption [16] [49] [50] [56] [59] [60] [61]. Environmental factors such as regulation, leadership backing, and informal networks also shape adoption decisions [6] [49] [50] [59] [62] [63].

This matters because AI tools are frequently marketed as easy productivity solutions, yet successful use often presumes hidden prerequisites: structured data, suitable workflows, basic AI literacy, time for experimentation, and access to technical help [16] [17] [18] [19]. The recurring pattern is that users are attracted by speed, convenience, and productivity gains, but hesitate because of concerns about accuracy, privacy, and control [12] [14] [36] [38] [53] [57].
Confidence level: High - multiple evidence streams converge on this tension [12] [38].

Discovery and Selection Barriers

Search Friction and Discovery Overload

The clearest finding is also an evidence gap: the literature does **not** directly study how non-technical users or SME owners search and shortlist AI tools in crowded marketplaces [1] [2] [3] [43] [53] [57]. Most retrieved studies are reviews or broad

References

1. Zejjari, I., Benhayoun, I. (2024) Bibliometric Analysis of Artificial Intelligence in Marketing: Revealing Publication Trends Using Bibliometrix R-Tool. *Contributions to Management Science*. https://doi.org/10.1007/978-3-031-67531-7_59
2. Riandhi, A.N., Arviansyah, M.R., Sondari, M.C. (2025) AI and consumer behavior: Trends, technologies, and future directions from a scopus-based systematic review. *Cogent Business and Management*. https://doi.org/10.1080/23311975.2025.2544984

3. Jo, H. (2023) Understanding AI tool engagement: A study of ChatGPT usage and word-of-mouth among university students and office workers. *Telematics and Informatics*. https://doi.org/10.1016/j.tele.2023.102067
4. Venkatesh, V., Morris, M.G., Davis, G.B., Davis, F.D. (2003) User acceptance of information technology: Toward a unified view. *MIS Quarterly: Management Information Systems*. https://doi.org/10.2307/30036540
5. Grandón, E.E., Pearson, J.M. (2004) Electronic commerce adoption: An empirical study of small and medium US businesses. *Information and Management*. https://doi.org/10.1016/j.im.2003.12.010
6. Booyse, D., Scheepers, C.B. (2024) Barriers to adopting automated organisational decision-making through the use of artificial intelligence. *Management Research Review*. https://doi.org/10.1108/MRR-09-2021-0701
7. Kučević, E., Leible, S., Lewandowski, T., (...), Ohlsen, F. (2024) A User-Based Study on the Acceptance of Artificial Intelligence-Based Decision-Support Systems. *Proceedings - 28th Pacific Asia Conference on Information Systems, PACIS 2024*. https://www.scopus.com/pages/publications/105026560175
8. Berger, E., Henke, J., Mehlstaubl, J., Paetzold-Byhain, K. (2026) Challenges hindering the application of GenAI methods in engineering design and the product development process: a meta-analysis. *Proceedings of the Design Society*. https://doi.org/10.1017/pds.2026.10576
9. Oomen, T., Ferreira Gonçalves, J., Mols, A. (2024) Rage Against the Artificial Intelligence? Understanding Contextuality of Algorithm Aversion and Appreciation. *International Journal of Communication*. https://www.scopus.com/pages/publications/85189476254
10. https://www.sciencedirect.com/science/article/pii/S266189072500376X
11. Almutairi, Waleed, Almatrodi, Ibrahim (2025) Trust Under Bounded Rationality: Exploring Human-AI Interaction in Decision-Making Through Large Language Models. *SAGE Open*. https://doi.org/10.1177/21582440251380135
12. Zubair, M., Alhassan, M.A., Bello, F. (2025) From Queries to Prompts: Comparing User Experience in Generative AI Tools and Search Engines. *BCS HCI 2025 - Human Centred Approaches and their Impact on AI System Design, Application, and Evaluation*. https://doi.org/10.14236/ewic/BCSHCI2025.13
13. Goethals, S., Martens, D., Evgeniou, T. (2022) The non-linear nature of the cost of comprehensibility. *Journal of Big Data*. https://doi.org/10.1186/s40537-022-00579-2
14. Oldemeyer, L., Jede, A., Teuteberg, F. (2025) Too Expensive or Worthwhile? Approaches to AI Assessment in SMEs. *Pacific Asia Conference on Information Systems*. https://www.scopus.com/pages/publications/105029268067

15. Ercan, S.E., Somers, G. (2025) Legislative Lag vs Technological Leap: Privacy and Data Protection Challenges when using AI-Driven Workplace Solutions. *Data Protection, Privacy and Artificial Intelligence: To Govern or To Be Governed,That is the Question*. https://www.scopus.com/pages/publications/105014575431

16. Shahzad, F., Aqeel, M., Arslan, A., Hurmelinna-Laukkanen, P. (2026) Exploring paradoxical barriers to AI adoption through the TOE framework. *Small Enterprise Research*. https://doi.org/10.1080/13215906.2026.2701701

17. Zavodna, L.S., Überwimmer, M., Frankus, E. (2024) Barriers to the implementation of artificial intelligence in small and medium-sized enterprises: Pilot study. *Journal of Economics and Management (Poland)*. https://doi.org/10.22367/jem.2024.46.13

18. Varun Krishna, M., Praveen, N., Kini, K. (2026) Empirical Analysis of AI Adoption in SMEs: Proposing a Strategic Framework and Readiness Scorecard. *IEEE International Conference on Emerging Synergy Science and Technology, ICESST 2026*. https://doi.org/10.1109/ICESST69086.2026.11582848

19. Waidringer, J., Giri, C., Beiker, S., Wittsten, J. (2026) A multiple case study of AI potential in small and medium sized companies (SME). *AIP Conference Proceedings*. https://doi.org/10.1063/5.0308483

20. Pérez-Campdesuñer, R., Sánchez-Rodríguez, A., García-Vidal, G., (...), De-Miguel-Guzmán, M. (2025) Artificial Intelligence in Ecuadorian SMEs: Drivers and Obstacles to Adoption. *Information (Switzerland)*. https://doi.org/10.3390/info16060443

21. Sánchez, E., Calderón, R., Herrera, F. (2025) Artificial Intelligence Adoption in SMEs: Survey Based on TOE–DOI Framework, Primary Methodology and Challenges. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app15126465

22. Chomutare, T., Tejedor, M., Svenning, T.O., (...), Ngo, P.D. (2022) Artificial Intelligence Implementation in Healthcare: A Theory-Based Scoping Review of Barriers and Facilitators. *International Journal of Environmental Research and Public Health*. https://doi.org/10.3390/ijerph192316359

23. Oldemeyer, L., Jede, A., Teuteberg, F. (2025) Investigation of artificial intelligence in SMEs: a systematic review of the state of the art and the main implementation challenges. *Management Review Quarterly*. https://doi.org/10.1007/s11301-024-00405-4

24. Schaffernak, Insa, Cecil, Julia, Kleine, Anne-Kathrin, Lermer, Eva (2025) Sociotechnical influences on the adoption and use of AI-enabled clinical decision support systems in ophthalmology: a theory-based interview study. *BMC Health Services Research*. https://doi.org/10.1186/s12913-025-13620-w

25. Rodríguez-Martín, Daniel, Pérez-López, Carlos (2024) Commercial symptom monitoring devices in Parkinson's disease: benefits, limitations, and trends. *Frontiers in Neurology*. https://doi.org/10.3389/fneur.2024.1470928

26. Dorta-González, P., López-Puig, A.J., Dorta-González, M.I., González-Betancor, S.M. (2024) Generative artificial intelligence usage by researchers at work: Effects of gender, career stage, type of workplace, and perceived barriers. *Telematics and Informatics*. https://doi.org/10.1016/j.tele.2024.102187

27. Nair, Monika, Svedberg, Petra, Larsson, Ingrid, Nygren, Jens M. (2024) A comprehensive overview of barriers and strategies for AI implementation in healthcare: Mixed-method design. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0305949

28. Boverhof, B.-J., Redekop, W.K., Visser, J.J., (...), Rutten-van Mölken, M.P.M.H. (2024) Broadening the HTA of medical AI: A review of the literature to inform a tailored approach. *Health Policy and Technology*. https://doi.org/10.1016/j.hlpt.2024.100868

29. Althoey, F., Sajjad, M., Houda, M., Waqar, A. (2025) Assessment of complexities in implementation of conversational AI for the digital transformation of small construction project. *Ain Shams Engineering Journal*. https://doi.org/10.1016/j.asej.2025.103370

30. Johnson, C., Egan, S.J., Carlbring, P., (...), Wade, T.D. (2024) Artificial intelligence as a virtual coach in a cognitive behavioural intervention for perfectionism in young people: A randomised feasibility trial. *Internet Interventions*. https://doi.org/10.1016/j.invent.2024.100795

31. Wenskovitch, John, Jefferson, Brett, Anderson, Alexander, (...), Fallon, Corey (2022) A Methodology for Evaluating Operator Usage of Machine Learning Recommendations for Power Grid Contingency Analysis. *Frontiers in Big Data*. https://doi.org/10.3389/fdata.2022.897295

32. Baumann, Steffen, Stone, Richard T., Abdelall, Esraa (2025) Smart Healthcare at Home: A Review of AI-Enabled Wearables and Diagnostics Through the Lens of the Pi-CON Methodology. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s25196067

33. Gorski, J., Rudloff, J., Ramgopal, S. (2025) Parental, Patient, and Provider Perspectives on the Use of Artificial Intelligence in Pediatric Care. *Academic Pediatrics*. https://doi.org/10.1016/j.acap.2025.102819

34. Selten, F., Klievink, B. (2024) Organizing public sector AI adoption: Navigating between separation and integration. *Government Information Quarterly*. https://doi.org/10.1016/j.giq.2023.101885

35. Ionaşcu, C.M. (2025) Artificial Intelligence Adoption in the European Union: A Data-Driven Cluster Analysis (2021–2024). *Economies*. https://doi.org/10.3390/economies13050145

36. Mach, M. (2025) AI Challenges From a Personal Data Perspective. *Pravnik*. https://www.scopus.com/pages/publications/105006712380

37. Quaranta, M., Amantea, I.A. (2025) The AI Act is Coming: Are E-Health Manufacturers Ready?. *Communications in Computer and Information Science*. https://doi.org/10.1007/978-3-031-74630-7_26

38. Olatunde, S., Shalan, A.M. (2025) Exploring the Privacy Paradox in AI Adoption: A Data-Driven Analysis of User Engagement. *2025 6th International Conference on Artificial Intelligence, Robotics, and Control, AIRC 2025*. https://doi.org/10.1109/AIRC64931.2025.11077547

39. Gong, Eun Jeong, Bang, Chang Seok, Lee, Jae Jun, (...), Kim, Jong Wook (2022) No-Code Platform-Based Deep-Learning Models for Prediction of Colorectal Polyp Histology from White-Light Endoscopy Images: Development and Performance Verification. *Journal of Personalized Medicine*. https://doi.org/10.3390/jpm12060963
40. Lee, Kyu-Hong, Lee, Ro-Woon, Yun, Jae-Sung, (...), Choi, Hyun-Seok (2024) Automated Diagnosis of Knee Osteoarthritis Using ResNet101 on a DEEP:PHI: Leveraging a No-Code AI Platform for Efficient and Accurate Medical Image Analysis. *Diagnostics*. https://doi.org/10.3390/diagnostics14212451
41. Mache, Stefanie, Bernburg, Monika, Würtenberger, Annika, Groneberg, David A. (2025) Artificial Intelligence in Primary Care: Support or Additional Burden on Physicians' Healthcare Work?—A Qualitative Study. *Clinics and Practice*. https://doi.org/10.3390/clinpract15080138
42. Gong, Eun Jeong, Bang, Chang Seok, Jung, Kyoungwon, (...), Lee, Jae Jun (2022) Deep-Learning for the Diagnosis of Esophageal Cancers and Precursor Lesions in Endoscopic Images: A Model Establishment and Nationwide Multicenter Performance Verification Study. *Journal of Personalized Medicine*. https://doi.org/10.3390/jpm12071052
43. Menaka, S., Selvam, V. (2024) Bibliometric Analysis of Artificial Intelligence on Consumer Purchase Intention in E-Retailing. *International Journal of Computational and Experimental Science and Engineering*. https://doi.org/10.22399/ijcesen.1007
44. Kim, T., Duhachek, A. (2023) Artificial Intelligence and Consumer Psychology. *The Cambridge Handbook of Consumer Psychology, Second Edition*. https://doi.org/10.1017/9781009243957.011
45. Ajzen, I. (1991) The theory of planned behavior. *Organizational Behavior and Human Decision Processes*. https://doi.org/10.1016/0749-5978(91)90020-T
46. Venkatesh, V., Davis, F.D. (2000) Theoretical extension of the Technology Acceptance Model: Four longitudinal field studies. *Management Science*. https://doi.org/10.1287/mnsc.46.2.186.11926
47. Trezza, D., de Luca Picione, G.L., Sergianni, C. (2025) AI Response Quality in Public Services: Temperature Settings and Contextual Factors. *Societies*. https://doi.org/10.3390/soc15050127
48. de Pretto, G., Bolisani, E., Nakash, M. (2025) AI for Knowledge-Driven Business Decisions: Capabilities and Constraints. *Proceedings of the European Conference on Knowledge Management, ECKM*. https://www.scopus.com/pages/publications/105020389216
49. Kunecová, J., Prester, J., Šebo, J., Palčič, I. (2026) Why do manufacturing firms struggle with artificial intelligence?. *Journal of Manufacturing Technology Management*. https://doi.org/10.1108/JMTM-05-2025-0432
50. Ledesma-Chaves, P., Gil Cordero, E., Navarro-García, A., Higuera Reina, J.A. (2026) Risk factors in the adoption of artificial intelligence by SMES: a comprehensive study. *European Journal of Innovation Management*. https://doi.org/10.1108/EJIM-06-2025-0719
51. Liu, J. (2025) AI in Automated and Remote UX Evaluation: A Systematic Review (2014–2024). *Advances in Human-Computer Interaction*. https://doi.org/10.1155/ahci/7442179

52. Alhitmi, H.K., Mardiah, A., Al-Sulaiti, K.I., Abbas, J. (2024) Data security and privacy concerns of AI-driven marketing in the context of economics and business field: an exploration into possible solutions. *Cogent Business and Management*. https://doi.org/10.1080/23311975.2024.2393743

53. Cevher, M.F. (2025) Examination of Consumer Perceptions Regarding Productive Artificial Intelligence Applications: A Qualitative Study on the Case of ChatGPT. *Romaya Journal: Researches on Multidisciplinary Approaches*. https://www.scopus.com/pages/publications/105004346864

54. Nordmann, K., Sauter, S., Stein, M., (...), Fischer, F. (2025) Evaluating the performance of artificial intelligence in summarizing pre-coded text to support evidence synthesis: a comparison between chatbots and humans. *BMC Medical Research Methodology*. https://doi.org/10.1186/s12874-025-02532-2

55. Weerheijm, B., Giest, S., Klievink, B. (2025) Complexity, understandability, and compatibility: A comparative study of AI advisory systems for National Security. *Government Information Quarterly*. https://doi.org/10.1016/j.giq.2025.102088

56. Harmon, J., Lazaros, E.J., Allam, H., (...), Truell, A.D. (2025) The growing adoption of artificial intelligence (AI) to drive innovation, resulting in enhanced competitiveness. *Issues in Information Systems*. https://doi.org/10.48009/3_iis_2025_2025_122

57. Strayhorn, T.L. (2025) College Students' Use and Perceptions of Artificial Intelligence (AI): A Survey Study. *Journal of College Student Development*. https://doi.org/10.1353/csd.2025.a962923

58. Tyagi, G., Verma, R.K., Sharma, A. (2025) SMEs and artificial intelligence in oman: A case of the insurance sector. *Issues in Information Systems*. https://doi.org/10.48009/3_iis_2025_138

59. Ganguly, B., Roy Chowdhury, S., Ghosh, R., Ghosh, D. (2025) What hinders AI adoption in MSMEs of India? – Importance and interlinkages between the barriers. *Indian Growth and Development Review*. https://doi.org/10.1108/IGDR-05-2024-0070

60. Rhouiri, M., Meyabe, M.H., Bensouda, M., Daidai, F. (2025) Behavioral and Financial Barriers to the Integration of Artificial Intelligence in the Tourism and Hospitality Industry: A Systematic Approach. *Behavioral Finance and AI Tools for Sustainability*. https://doi.org/10.4018/979-8-3693-9684-1.ch003

61. Sathyanarayana, N., Varanasi, V., Fathima, Y., (...), V, V.L. (2026) Cost and Accessibility of AI: Breaking Barriers for MSMEs. *Opportunities and Challenges of AI in Micro, Small, and Medium-Sized Enterprises*. https://doi.org/10.4018/979-8-3373-2817-1.ch004

62. Schoeman, F., Seymour, L.F. (2022) Understanding the Low Adoption of AI in South African Medium Sized Organisations. *EPiC Series in Computing*. https://doi.org/10.29007/c4rr

63. Malarout, N., Shenoy, S.S., Shetty, D.K., (...), Almeida, R.S. (2026) Artificial intelligence in entrepreneurship: a comprehensive review of adoption patterns, challenges, and strategic implementation in early-stage ventures. *Connecting Intelligence: Trends in Computation and Data Communication*. https://doi.org/10.1201/9781003773504-123